Reg. No. | | | | | | | | | | | [14CS/IS46]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2016

DESIGN AND ANALYSIS OF ALGORITHMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Instructions: Answer any FIVE full questions choosing at least TWO questions from each Part.
2. Any missing data may be suitably assumed

PART – A

1. a. With the help of a flow diagram, explain the various types of algorithm design and analysis process. **08**

b. Discuss any six important problem types. Give examples for each. **06**

2. a. Compare the order of growth of $\frac{1}{2}n(n-1)\ and\ n^2$. **03**

b. List and explain basic asymptotic efficiency classes. **07**

c. Define and explain with example Big-Oh. **04**

3. a. Sort the following list 66, 55,43,32,21 using selection sort in an ending order. Show that its complexity is $O(n^2)$. **07**

b. Explain the brute force string matching algorithm with an example. **07**

4. a. Outline an exhaustive search algorithm to solve the traveling salesman problem. **04**

b. State the merge sort algorithm and analyse its complexity. **05**

c. Trace the quick sort algorithm for the following data set 89, 45,68,90,29. **05**

PART – B

5. a. Define:

i. Graphs

ii. Weighted graphs

iii. Directed graph

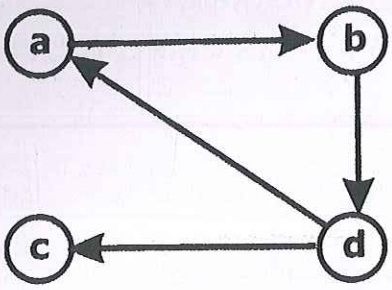
iv. Connected graph **04**

b. Design an algorithm for DFS. How is it different from BFS with suitable example? **10**

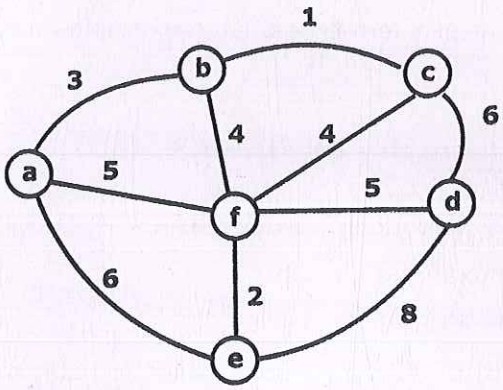
6. a. Design an algorithm for solving all pair shortest path problem. Find shortest path for the given graph represented is adjacency matrix. **08**

$$\begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & \infty & 3 & \infty \\ 2 & 0 & \infty & \infty \\ \infty & 7 & 0 & 1 \\ b & \infty & \infty & 0 \end{bmatrix} \end{array}$$

b. What is dynamic programming? Explain Warshall algorithm to find the transitive closer of a directed graph. Apply this algorithm to the graph given below: **07**



7. **a.** Apply Kruskai's algorithm to find the minimum spanning types of the following graph. **07**



b. Construct a Huffman code for the following table: **07**

Char	A	B	C	D	E
Probability	0.4	0.1	0.15	0.2	0.15

8. **a.** Discuss briefly knapsack problem using brute force method. **04**

b. Write an algorithm for string matching using Robin Karp algorithm also provide worst-case complexity. **10**